

Quantum Cryptography and the BB84 Protocol

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1 Introduction

Cryptography can be defined as the act of using algorithms to code and encrypt data during transmission to ensure that only authorised people can read it. Before evaluating quantum mechanics in cryptography, it is essential to analyse the fundamental limitations found in classical private and public key methods.

1.1 Private key encryption

Private key cryptography utilises a shared private key for both encryption and decryption. One algorithm commonly used is AES (Advanced Encryption Standard), which operates as a block cipher [1]. The AES algorithm splits input data into 128-bit blocks and outputs 128 bits of encrypted cipher text [1]. To achieve this, the cipher arranges each block into a 4x4 matrix grid of bytes [1]. A series of four linked operations (SubBytes, ShiftRows, MixColumns, and AddRound-Key) are executed across multiple rounds depending on the key size to encrypt the plaintext [1]. To decrypt the data, these exact stages are executed in reverse [1]. Because of its efficiency, AES is widely implemented in wireless security to protect Wi-Fi networks and is standardly used to securely hash stored passwords instead of leaving them in plaintext [1]. However, private key cryptography has its limitations. Firstly, it relies on a shared key, which needs to be transmitted securely at the start. If this shared key were to be compromised during the initial transmission, all communication would be compromised. Secondly, it is vulnerable to brute-forcing. As quantum computers and processing capabilities become more mainstream in the future, advanced computing power could eventually be leveraged to crack these cryptographic keys.

1.2 Public key encryption

Public key cryptography makes use of 2 keys; a public key to encrypt the data and a private key which is kept secret by the recipient to decrypt the data [2]. It is more secure than private key encryption since there are 2 different keys, and it allows communications over untrusted networks. One example of an algorithm used is RSA (Rivest-Shamir-Adleman). This algorithm relies on the difficulty in

factoring large prime numbers [3]. To generate a key, 2 distinct prime numbers p and q are selected [3].

Prime numbers: p, q

Next, the modulus n is calculated [3].

$$n = p \times q$$

Euler's totient function is calculated using the formula:

$$\phi(n) = (p - 1)(q - 1)$$

An integer e is chosen as the public exponent where

$$1 < e < \phi(n)$$

and the greatest common divisor is given by [3]:

$$\gcd(e, \phi(n)) = 1$$

The decryption exponent d is computed where [3]:

$$(d \times e) \equiv 1 \pmod{\phi(n)}$$

The public key is represented as:

$$(n, e)$$

and the private key is given by [3]:

$$(n, d)$$

When encrypting data (a) it is first converted in a numerical representation and then the ciphertext text C is calculated using the public key and the formula:

$$C = a^e \pmod{n}$$

When decrypting data the private key and the following formula are used:

$$a = C^d \pmod{n}$$

RSA is an extremely strong algorithm and is used to encrypt web traffic, and is also used in secure banking. However, it has its downsides. It is slow when dealing with large amounts of data, due to the complex encryption process and it can also be decrypted relatively quickly with quantum computing and Shor's algorithm, which can find the prime factors of large numbers in a short amount of time.

2 Theoretical Principles of Quantum Mechanics

2.1 The Quantization of Light and Photon Energy

In classical physics, we consider light to be a wave on the electromagnetic spectrum, but in quantum mechanics we consider light to be quantised into discrete packets of energy, called photons. This was first discovered by Max Planck in 1900, explaining this theory of black body radiation. It was then expanded upon by Albert Einstein in 1905, which led to him winning a Nobel Prize for the photoelectric effect, where light above a threshold frequency caused electrons to be emitted from the surface of the metal. This couldn't be explained by the classical wave theory of light so, it led to the quantisation of light.[4] The equation which Max Planck created was [5] :

$$E = hf$$

The energy of a single photon E (joules) is given by the product of the frequency of the light wave (hertz) and Planck's constant h .

2.2 Polarisation in transverse waves

Polarisation is a property of transverse waves, where the oscillations of the electric field vector are restricted to a single plane, perpendicular to the direction of wave propagation. Light is a transverse wave on the EM spectrum, and is unpolarised, which means it oscillates in every possible plane perpendicular to the direction of propagation. When unpolarised light passes, through a polarising filter, oscillations perpendicular to the transmission axis are blocked while oscillations parallel to the transmission axis pass through. In quantum cryptography, this physical behaviour is utilised by implementing two orthogonal bases.[6] The first basis is the canonical basis where the polarisation can either be 0° (horizontal) or 90° (vertical). The second basis is the diagonal basis where the polarisation can be 45° (diagonal-left) or 135° (diagonal-right).

2.3 Wavefunction collapse and Malus's Law

In classical optics, when a polarised beam of light hits another polarising filter at angle θ , the intensity of the light can be calculated using the formula

$$I = I_0 \cos^2 \theta$$

where the final intensity I is equal to the initial intensity I_0 multiplied by $\cos^2(\theta)$ where θ is the angle between the the transmission axis of the polariser and polarization axis of the incident light. In quantum mechanics, a photon's intensity can't be reduced, since there are only 2 possible outcomes; the photon is either absorbed or passes through. Therefore, rather than measuring intensity, it measures probability P of a photon passing through using the formula.

$$P = \cos^2 \theta$$

As per the Copenhagen interpretation of quantum mechanics, physical systems don't have definite properties before measurement, and quantum mechanics can only predict the probabilities that measurements will produce certain results.[7] This quantum state is represented by a wavefunction ψ where the probability of a measurement producing a certain result is given by the squared magnitude of the wavefunction $|\psi|^2$, since probability is always positive. Measuring the probability affects the system causing the set of probabilities to reduce to one value after immediately after the measurement. This act is known as wavefunction collapse when the photon is forced from its superposition of states into a distinct 'eigenstate'. One example of this is sending a canonical photon with a (0°) polarisation, through a diagonal filter (45°) . After passing through, its quantum state instantly collapses and it is now a 45° photon, with its old state completely erased.[6], [7]

2.4 No-cloning theorem

In 1982 Wootters, Zurek, and Dieks worked together to come up with the no-cloning theorem in quantum mechanics which states that because quantum operations are unitary linear transformations of quantum states, arbitrary unknown states cannot be perfectly copied.[8] In order to copy a quantum state it measured. However, according to the Copenhagen interpretation, measurement will cause the wavefunction to collapse, if using a different basis.[7] Therefore, it is not possible to read the state of a photon without modifying the original state.

3 The BB84 protocol

Quantum cryptography relies on the property of measurements to alter the quantum state that they are used on [9]. The first quantum key distribution BB84 was created by Charles Bennett and Gilles Brassard in 1984. It works in the following way. There are 2 people Alice (the sender) and Bob (the receiver) who wish to communicate.

1. Alice randomly chooses the value of a bit to be sent (0 or 1).
2. Alice randomly chooses the encoding basis (canonical or diagonal).
3. Alice generates a photon and encodes the bit in the same polarisation as the photon and sends it to Bob along the quantum channel such as a fiber optic cable.
4. Bob randomly chooses a measurement basis (canonical or diagonal), since he does not know what encoding basis Alice has chosen.
5. Bob will measure the arriving photon. If his measurement basis matches Alice's encoding basis, he will detect the same bit as the one Alice sent. If he selects the other basis, he will detect a random bit value, since the photon has collapsed to a different state and the original state is completely erased.

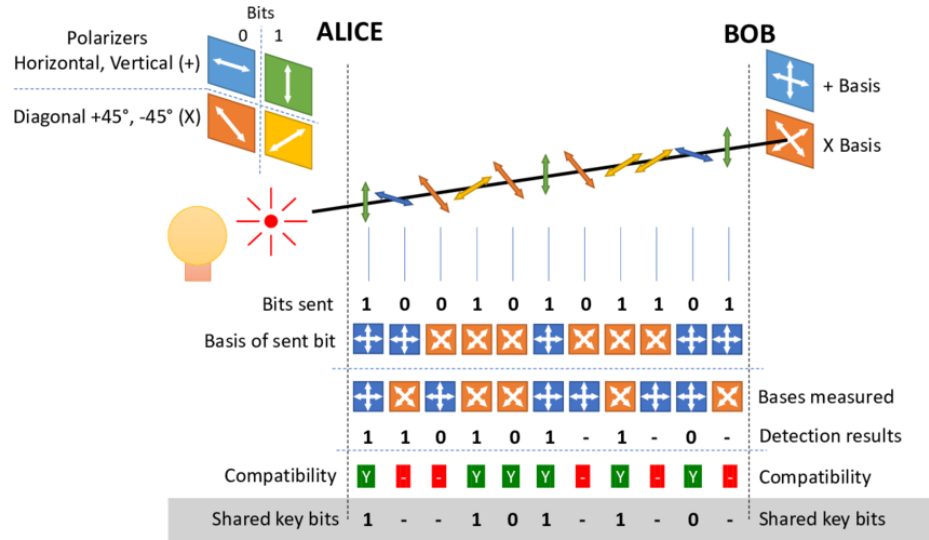


Figure 1: A visual explanation of the BB84 protocol [10].

This is repeated many times, and Alice and Bob both keep records of the bases used, states sent and detected and the timing. After collecting enough records, they inform each other of their choice of basis for each photon, but not the bit values sent or measured. They do this on a public unencrypted channel such as the internet. They discard data for events where the bases used do not match or when the photon has been lost. This results in around 50% of the data being discarded since Bob has a 50% chance to guess the correct encoding basis used by Alice. As a result, they now have a string of identical bits, which can be used as a one time pad in a private key protocol. Above is an image visually explaining the theory, resulting in the shared key bit which is the secret key.

3.1 Classical cryptography

Alice and Bob can now use that key to securely communicate together. Alice can now apply an XOR operation to every bit of her message and the secret key generated using the BB84 protocol, resulting in an encrypted message. It can only be decrypted by applying an XOR operation by Bob, who has the same secret key. Below is an explanation of the operations performed to encrypt the message into cipher text and decrypt the message into plaintext using the secret key.

4

Eavesdropping and Mismatch Probabilities

USING EXCLUSIVE OR (XOR) IN CRYPTOGRAPHY			
XOR LOGIC	0 XOR 0 = 0	Same Bits	
	1 XOR 1 = 0	Same Bits	
	1 XOR 0 = 1	Different Bits	
XOR Symbol $\oplus$	0 XOR 1 = 1	Different Bits	
ENCRYPT			
	0 0 1 1 0 1 0 1	Plaintext	
$\oplus$	1 1 1 0 0 0 1 1	Secret Key	
	= 1 1 0 1 0 1 1 0	Ciphertext	
DECRYPT			
	1 1 0 1 0 1 1 0	Ciphertext	
$\oplus$	1 1 1 0 0 0 1 1	Secret Key	
	= 0 0 1 1 0 1 0 1	Plaintext	

Figure 2: A visual explanation of the encryption and decryption process [11].

4.1 Interception

There now exists an eavesdropper (Eve) who wishes to find out the secret key used to decrypt Alice and Bob's messages. Due to the no-cloning theorem, she cannot copy the photons, so her only option is to intercept the photon, measure it using a random basis, and send a new photon in the basis she just measured.

4.2 Malus's Law and Induced State Collapses

Since Eve does not know what bases Alice and Bob are using, she has a 50% chance of selecting the correct basis. If she chooses the correct basis, no error is introduced in the process. However, if she chooses the wrong basis, her measurement of the photon will cause the wavefunction to collapse, completely changing the state of the photon. If Alice sends a canonical photon (0°), and Eve measures it diagonally, the polarisation is forced to (45°). Eve then sends this to Bob who measures it in the canonical basis. According to Malus's Law :

$$P = \cos^2(45^\circ) = 0.5$$

As a result, Bob has a 50% of receiving a 1 or 0. Therefore, due to Eve's interference Bob has a 50% chance of receiving the wrong bit, even when using the same basis as Alice.

4.3 Quantum Bit Error Rate (QBER) and detection

The total Quantum Bit Error Rate (QBER) introduced into Alice and Bob's secret key by an intercepting eavesdropper is calculated as:

$$\text{QBER} = P(\text{Eve Error}) \times P(\text{Bob Error} \mid \text{Eve Error})$$

$$\text{QBER} = 0.5 \times 0.5 = 0.25$$

Therefore, the presence of Eve introduces an unavoidable 25% error rate into the final secret key. In order to check if there is an eavesdropper listening into their transmission, Alice and Bob compare a small part of the secret key. If the error rate is close to 25%, it is obvious that the eavesdropper is listening, so they will discard the key. Otherwise, they will use the rest of the secret key bit string as a one-time pad to encrypt and decrypt their messages.

5 Engineering Challenges and the Future

5.1 Physical Infrastructure Limitations

Quantum key distribution is theoretically secure since it is not possible to contravene the laws of physics, but there are a few practical issues. Firstly, it is susceptible to a MITM (man-in-the-middle) attack, where an attacker such as Eve could act as Bob to Alice and Alice to Bob, by inserting herself between the 2 people. A MITM attack allows Eve to intercept and relay messages between Alice and Bob. To prevent this, real-world QKD systems require pre-existing classical authentication mechanisms to ensure Alice and Bob are communicating directly rather than via an untrusted intermediary.[12]

Secondly, designing and building equipment which can produce and detect singular photons over long distances is difficult. Therefore, the protocol is implemented using a source that produces multiple photons. This has consequences, since Eve can now isolate one of the photons while allowing the rest to pass through unchanged. She can store the photon in a quantum memory for later, when the bases are shared on the classical unencrypted channel, to finally measure it in the correct basis. This enables Eve to obtain information without disturbing the photons received by Bob.[12]

Finally, there are difficulties in transmitting photons over long distances using optical fibre cable, since the signal attenuates exponentially with transmission distance. For example, some photons may be absorbed into the fibre or reflected backwards. One solution to this is a repeater, which amplifies the signal by measuring the original signal, copying it and transmitting it with greater power. However, this can't be applied to quantum communication due to the

no-cloning theorem. Measuring the incoming signal would cause the state of the photon to collapse, erasing the previous state.[12]

5.2 Quantum Key Distribution vs. Post-Quantum Cryptography

As the deployment of viable quantum computers running Shor’s algorithm approaches, securely transmitting data requires choosing between 2 methods QKD (Quantum Key Distribution) and PQC (Post-Quantum Cryptography).[13] PQC relies on mathematical complexity, utilising algorithms like lattice based cryptography that is too computationally intensive for classical and quantum computers. Unlike QKD, PQC is driven by software and does not require any hardware such as photon emitters and detectors.[13] QKD’s security is guaranteed by the laws of physics whereas PQC relies on mathematical assumptions. PQC also does not have of the physical limitations mentioned above, since it is implemented digitally.[13]

6 Conclusion

Quantum cryptography has distinct advantages over classical cryptography since it relies on the fundamental laws of physics rather than mathematical complexity. As a result of the no-cloning theorem, wavefunction collapse, and Malus’s Law, an unavoidable physical disturbance is introduced by any interception, enabling definitive eavesdropper detection. Although theoretically secure, its real-world application remains constrained by current hardware limitations such as the difficulty of generating ideal single photons, and restricted communication ranges caused by exponential fiber-optic signal attenuation. As the development of viable quantum computers approaches reality, the threat of Shor’s algorithm to global security grows increasingly urgent. In response, QKD and PQC should not be viewed as rivals, but rather as complementary technologies with distinct use cases. PQC is highly scalable over existing legacy infrastructure, making it ideal for widespread consumer applications, whereas QKD provides an information-theoretically secure network capable of withstanding any future computational breakthroughs.

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